$^{1}H(^{34}S,^{35}Cl\gamma)$ **2021Lo09**

$^{34}S(p,\gamma)^{35}Cl$ in inverse kinematics.

2021Lo09: ^{34}S was produced by the off-line ion source (OLIS) and accelerated by the ISAC accelerator at TRIUMF. $10^{10}-10^{11}$ pps 7^{+} ^{34}S ions were delivered to a windowless 2-10 Torr hydrogen gas target with an effective length of 12.3 cm. Elastically scattered protons were detected by two silicon surface barrier detectors inside the gas target. Prompt γ rays from the compound recoil nuclei were detected by a BGO array in coincidence with the heavy-ion recoils separated by the DRAGON separator and identified by two MCP detectors, a DSSD, and an ionization chamber at the focal plane. Deduced resonance strengths ($\omega\gamma$) for 8 resonances and the $^{34}S(p,\gamma)^{35}Cl$ astrophysical reaction rate.

^{35}Cl Levels

Values of resonance strength $\omega\gamma$ under comments are from direct measurements in 2021Lo09.

E(level)†	Comments
6643	E_R=272 keV. $\omega\gamma=1.22\times10^{-5}$ eV *36*.
6674	E_R=303 keV. $\omega\gamma \le 7.2\times10^{-6}$ eV.
6761	E_R=390 keV. $\omega\gamma \le 6.9\times10^{-6}$ eV.
6778	E_R=407 keV. $\omega\gamma=8.0\times10^{-4}$ eV *19*.
6802	E_R=431 keV. $\omega\gamma \le 3.9\times10^{-5}$ eV.
6823	E_R=452 keV. $\omega\gamma \le 3.1\times10^{-5}$ eV.
6842	E_R=471 keV. $\omega\gamma=9.4\times10^{-5}$ eV *29*.
6866	E_R=496 keV. $\omega\gamma=1.37\times10^{-2}$ eV *22*.

† Excitation energies are deduced by evaluators from resonance energies $E_R+S_p(^{35}Cl)$, where $S_p(^{35}Cl)$=6370.81 *4* (2021Wa16).